GATE CSE 2027

The PYQ Playbook

Every official GATE CS paper from 2010 to 2026, broken down by subject and chapter: where the marks have been, how steady each subject is, and what to study first.

27	1,755	11	65
papers analysed 2010-2026	questions classified	technical subjects + General Aptitude	chapters tracked paper by paper

SUBJECTS BY RECENT AVERAGE (2024-26 papers, marks out of 85, plus 15 for aptitude)

01	Programming & Data Structures	10.3
02	Operating Systems	9.0
03	Computer Networks	8.2
04	Computer Organization & Architecture	8.2
05	Engineering Mathematics	7.8
06	Databases	7.5
07	Algorithms	7.3
08	Digital Logic	7.3
09	Theory of Computation	6.8
10	Compiler Design	6.7
11	Discrete Mathematics	5.8
+	General Aptitude	15

How to use this playbook

Built for revision. Learn each subject from your books or lectures first; use these pages to decide what to study first and to revise from.

WHILE LEARNING
Read the subject sheet
When you finish a subject, read its two pages. The chapter grid shows which parts GATE keeps coming back to.

DURING REVISION
Summary and traps
Ten minutes per subject: read the summary and the "watch for" list, then practise the chapters with the most questions.

LAST TWO WEEKS
Reliability and checklist
Section 06 ranks every chapter by how reliably it is asked. Section 07 is the checklist to run through before the exam.

What's inside

01	**Where the marks are**	Subjects ranked by recent average, with tiers
02	**Marks per paper**	Every subject's marks in all 27 papers
03	**Trends and stability**	Era averages, spread, highest and lowest
04	**Paper format**	Structure, negative marking, question types
05	**Subject sheets**	Two pages per subject: marks, summary, chapter grid
06	**Chapter reliability**	How often each chapter appeared in the last 13 papers
07	**Exam-day checklist**	Habits to run on autopilot during the paper
08	**About the data**	Sources, method and what is excluded

Reading the labels

Tier 1 · Critical
The three subjects with the highest recent average.

Tier 2 · High
The next four, each worth 7.3-8.2 marks a paper recently.

Tier 3 · Steady
Fewer marks, but rarely skipped. Not optional.

Recent avg
Average marks in the 6 papers of 2024-2026, the best guide to 2027.

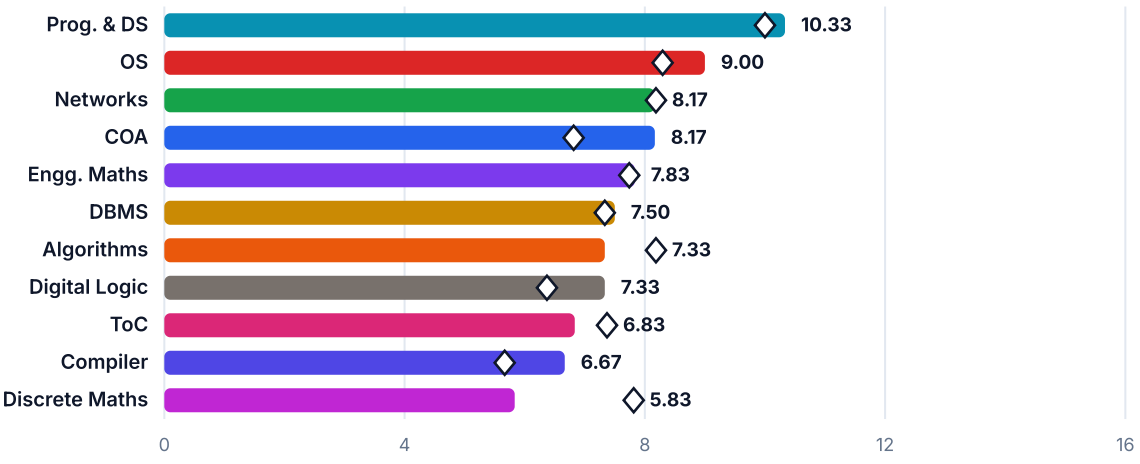
Scope. Marks are counted by each question's main subject in the ExamPrep question bank. Software engineering and web technology questions (asked only up to 2016) are shown separately and are not part of any average.

01 Where the marks are

Subjects ranked by their **recent average**: marks per paper over the six papers of 2024-2026. The diamond marks the 27-paper average. Technical marks total 85 per paper.

#	SUBJECT	RECENT AVG 2024-26	SHARE of 85	27-PAPER avg	PRIORITY
1	Programming & Data Structures	**10.33**	12.2%	10.00	Tier 1 · Critical
2	Operating Systems	**9.00**	10.6%	8.30	Tier 1 · Critical
3	Computer Networks	**8.17**	9.6%	8.19	Tier 1 · Critical
4	Computer Organization & Architecture	**8.17**	9.6%	6.81	Tier 2 · High
5	Engineering Mathematics	**7.83**	9.2%	7.74	Tier 2 · High
6	Databases	**7.50**	8.8%	7.33	Tier 2 · High
7	Algorithms	**7.33**	8.6%	8.19	Tier 2 · High
8	Digital Logic	**7.33**	8.6%	6.37	Tier 3 · Steady
9	Theory of Computation	**6.83**	8.0%	7.37	Tier 3 · Steady
10	Compiler Design	**6.67**	7.8%	5.67	Tier 3 · Steady
11	Discrete Mathematics	**5.83**	6.9%	7.81	Tier 3 · Steady
	Total technical	**85.00**	**100%**		
+	General Aptitude (separate 15 marks)	**15.00**	-	15.00	Fixed 15 marks

Recent average vs 27-paper average



27.5

Marks a paper from the three Tier 1 subjects (Prog. & DS, OS, Networks) in 2024-26: 32.4% of the technical marks.

19.3

Marks from the three lowest subjects together. Still more than any single Tier 2 subject, so don't skip them.

02 Marks per paper, 2010-2026

Each cell is a subject's marks in one paper. Shading is relative to that subject's own range, so a dark cell is a big year *for that subject*.

2010-2017 · 14 papers

	'10	'11	'12	'13	'14 S1	'14 S2	'14 S3	'15 S1	'15 S2	'15 S3	'16 S1	'16 S2	'17 S1	'17 S2
Prog. & DS	10	8	10	7	3	11	8	12	8	14	11	11	11	13
OS	7	7	9	10	8	7	7	10	8	6	9	9	6	6
Algorithms	7	9	9	7	13	8	9	8	8	11	9	10	6	8
Networks	8	6	9	7	9	9	9	8	9	8	10	9	8	7
Discrete Maths	7	3	9	8	13	9	13	10	12	5	8	10	7	5
Engg. Maths	8	10	7	5	10	10	9	9	9	10	5	4	8	10
ToC	7	8	5	8	6	6	6	5	7	5	9	9	10	9
DBMS	7	7	11	7	8	8	8	6	6	6	5	6	8	8
COA	7	8	5	7	7	7	5	3	5	5	5	9	10	6
Digital Logic	7	7	5	5	4	5	6	5	5	7	7	3	5	9
Compiler	6	6	6	7	3	4	4	5	4	3	7	5	6	4
Software Engg.	4	6		7	1	1	1	4	4	5				
Technical total	85	85	85	85	85	85	85	85	85	85	85	85	85	85
Aptitude	15	15	15	15	15	15	15	15	15	15	15	15	15	15

2018-2026 · 13 papers

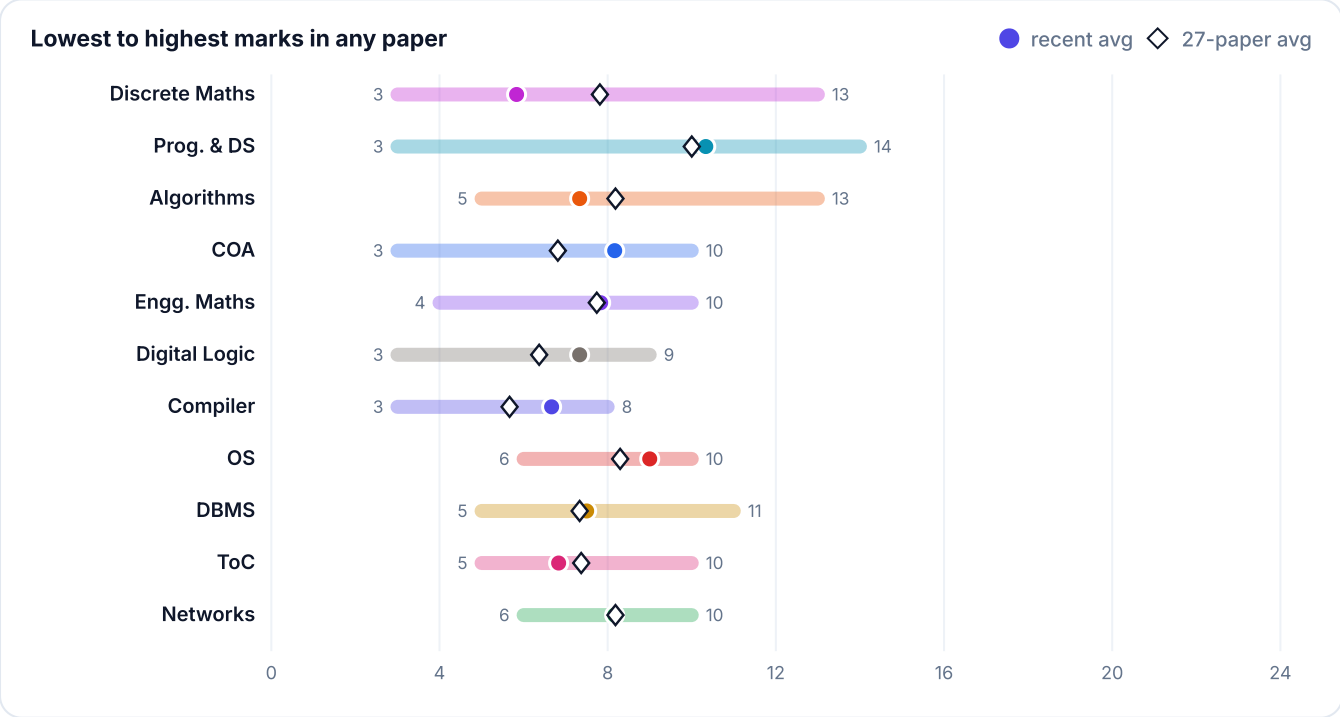
	'18	'19	'20	'21 S1	'21 S2	'22	'23	'24 S1	'24 S2	'25 S1	'25 S2	'26 S1	'26 S2
Prog. & DS	12	14	12	6	8	10	9	8	8	11	12	12	11
OS	9	10	10	6	7	10	9	10	10	8	7	10	9
Algorithms	8	6	8	12	10	5	6	7	6	7	8	7	9
Networks	7	9	6	9	7	10	8	9	9	8	6	8	9
Discrete Maths	9	6	8	7	4	13	10	6	9	6	3	7	4
Engg. Maths	7	7	5	8	9	6	6	8	6	7	9	8	9
ToC	8	8	9	8	9	8	8	7	7	8	7	6	6
DBMS	6	8	8	10	8	7	5	8	8	8	9	6	6
COA	8	3	9	5	6	7	8	8	8	8	9	7	9
Digital Logic	6	8	6	6	9	5	8	6	6	8	9	8	7
Compiler	5	6	4	8	8	4	8	8	8	6	6	6	6
Technical total	85	85	85	85	85	85	85	85	85	85	85	85	85
Aptitude	15	15	15	15	15	15	15	15	15	15	15	15	15

Why some early totals include Software Engineering. Until 2016 a few questions per paper covered software engineering and web technologies. They are shown in their own row and left out of every subject average, because they are no longer in the syllabus.

03 Trends and stability

Three eras of the exam. The direction compares 2024-26 with 2018-23: a change under 0.75 marks counts as flat.

SUBJECT	2010-17 14 papers	2018-23 7 papers	2024-26 6 papers	DIRECTION	STD. DEV.	SPREAD
Prog. & DS	9.79	10.14	**10.33**	→ Flat	2.48	Swings
OS	7.79	8.71	**9.00**	→ Flat	1.46	Steady
COA	6.36	6.57	**8.17**	↑ Rising	1.81	Steady
Networks	8.29	8.00	**8.17**	→ Flat	1.12	Steady
Engg. Maths	8.14	6.86	**7.83**	↑ Rising	1.76	Steady
DBMS	7.21	7.43	**7.50**	→ Flat	1.39	Steady
Algorithms	8.71	7.86	**7.33**	→ Flat	1.85	Steady
Digital Logic	5.71	6.86	**7.33**	→ Flat	1.54	Steady
ToC	7.14	8.29	**6.83**	↓ Falling	1.36	Steady
Compiler	5.00	6.14	**6.67**	→ Flat	1.54	Steady
Discrete Maths	8.50	8.14	**5.83**	↓ Falling	2.87	Swings



Swings

Discrete Maths, Prog. & DS move by several marks from paper to paper (std. dev. 2 or more). A light year doesn't mean the next one will be.

Steady

Engg. Maths, COA, Algorithms, Networks, DBMS, OS, ToC, Compiler, Digital Logic stay within a narrow band. Their averages are reliable predictions for 2027.

04 Paper format

Not a topic, but it changes how you attempt the paper. Every number below is read from the papers themselves.

15
General Aptitude
10 questions: 5 × 1 mark + 5 × 2 marks

25
Technical, 1 mark
25 questions × 1

60
Technical, 2 marks
30 questions × 2

65 questions, 100 marks, 3 hours. The same structure in all 27 papers since 2010.

Negative marking (2024-26 papers)

TYPE	WRONG ANSWER
MCQ · 1 mark	**-1/3**
MCQ · 2 marks	**-2/3**
MSQ (select all correct)	**None**
NAT (numerical answer)	**None**

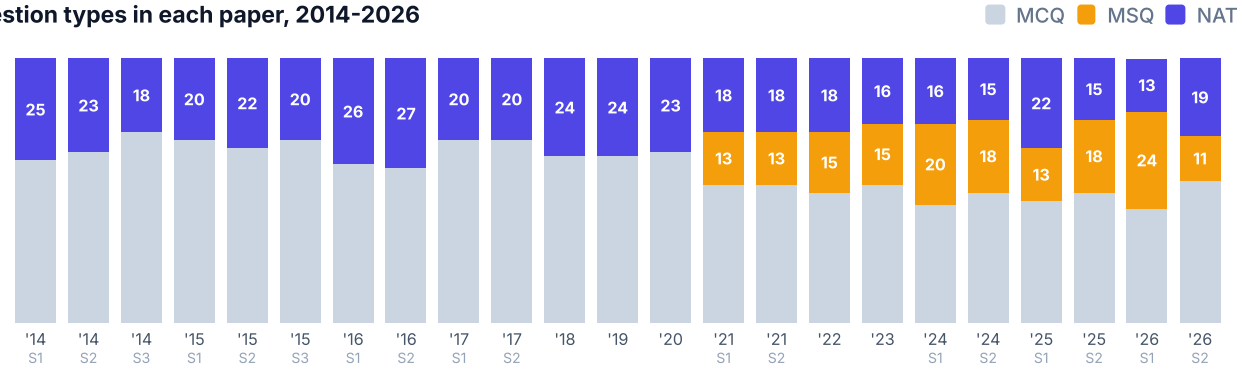
MSQs give no partial credit: every correct option must be chosen. In 2010-2013, two-mark linked questions sometimes carried no penalty on the second part.

Where numerical questions sit

PAPER	NAT · 1 MARK	NAT · 2 MARKS
2023	6	10
2024 S1	4	12
2024 S2	2	13
2025 S1	7	15
2025 S2	4	11
2026 S1	4	9
2026 S2	6	13

Numerical questions still appear in both blocks (2026 S1: 4 + 9; 2026 S2: 6 + 13), and most of them are two-markers.

Question types in each paper, 2014-2026



Practical rule. MSQs arrived in 2021 and NATs in 2014; together they are now 34 of 65 questions with no penalty. Answer all of them. Guess on an MCQ only after removing an option.

SUBJECT SHEET 1 OF 12

Programming & Data Structures

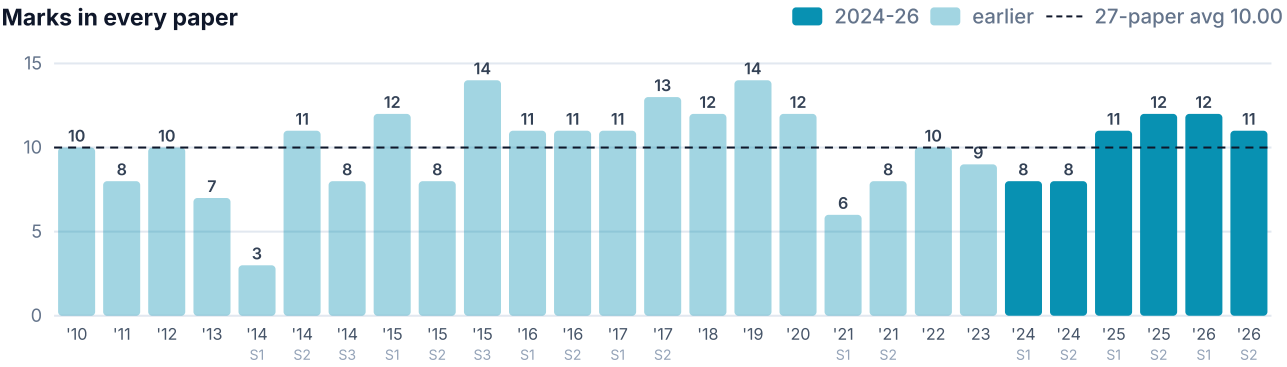
Tier 1 · Critical | Recent avg 10.33 | Asked in 27/27 papers | Last asked 2026 S2

IN SHORT

- → C code tracing is the largest single topic: pointers, recursion, static variables, parameter passing and printf output.
- → Trees are next: binary search tree properties, traversals, AVL heights and heaps.
- → Stacks, queues, linked lists, arrays and hashing appear as short "what is the state after these operations" questions.
- → Keep a written table of every variable as you trace. Tracing in your head is where marks go.

RECENT AVG · 2024-26	27-PAPER AVG	LOWEST	HIGHEST
10.33	10.00	3	14
12.2% of 85	2010-2026	'14 S1	'15 S3, '19

Marks in every paper



2010-17	2018-23	2024-26
9.79	10.14	10.33

→ Flat · std. dev. 2.48 · recent questions: 14 MCQ, 9 MSQ, 17 NAT (18 one-mark, 22 two-mark)

1 Static and global variables

They keep their value across calls. Recursion questions lean on this.

2 Pre vs post increment

Inside expressions and array indices, the order of evaluation decides the answer.

3 Heap index arithmetic

Check whether the array is 0- or 1-indexed before using the child formula.

Prog. & DS · chapters and years asked

Questions per chapter in each of the last 13 papers. A question tagged with two chapters counts in both. The right-hand columns compare with 2010-17.

Chapter	'18	'19	'20	'21 S1	'21 S2	'22	'23	'24 S1	'24 S2	'25 S1	'25 S2	'26 S1	'26 S2	2018-26	2010-17
Programming Languages	5	6	3	1	4	3	2	3	3	3	4	3	4	13/13	14/14
Trees	2	2	4	1	2	1	3	2	1	2	4	4	2	13/13	14/14
Arrays		3	1	1	1	2			4	1	2			8/13	9/14
Graphs	1			1	1		1	2		1	2	2		8/13	6/14
Linked Lists	1		1		1	1	1			1	1	1		8/13	4/14
Stacks & Queues	1			1		1	2		1		1		1	7/13	8/14
Hashing			1			1	1			1		1	1	6/13	5/14

Most-asked topics since 2018

TOPIC	QUESTIONS	PAPERS
Programming In C	39	13/13
Array	15	8/13
Recursion	12	9/13
Binary Search Tree	12	9/13
Binary Heap	11	9/13
Linked List	8	8/13
Tree Traversal	7	7/13
Binary Tree	6	6/13

STUDY ORDER C programming → Trees & heaps → Linear structures → Hashing

SUBJECT SHEET 2 OF 12

Operating Systems

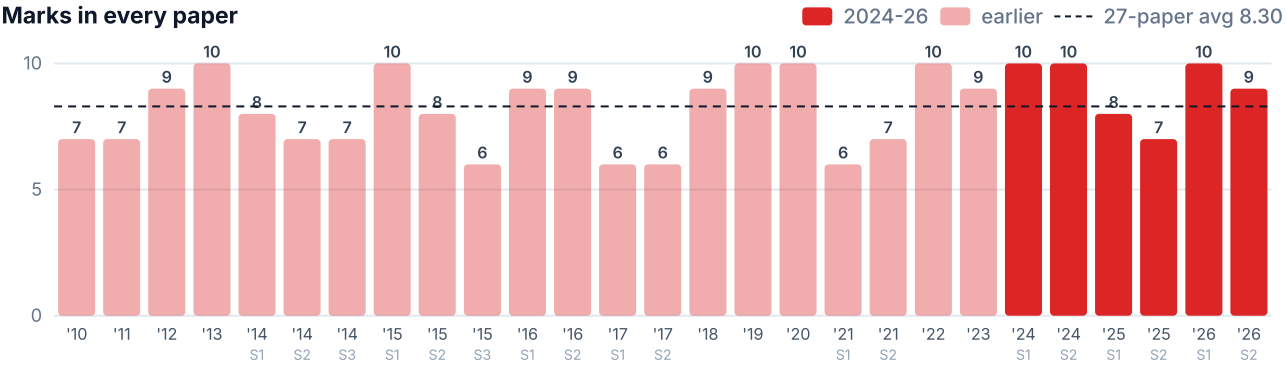
Tier 1 · Critical | Recent avg 9.00 | Asked in 27/27 papers | Last asked 2026 S2

IN SHORT

- → CPU scheduling: draw the Gantt chart every time, then read off waiting and turnaround times.
- → Synchronization: semaphores and critical sections. Test each claim with a concrete interleaving.
- → Memory management: paging, TLB effective access time, and page replacement (FIFO, LRU, optimal).
- → Deadlock: the four conditions and the Banker's safety check.

RECENT AVG · 2024-26	27-PAPER AVG	LOWEST	HIGHEST
9.00	8.30	6	10
10.6% of 85	2010-2026	'15 S3, '17 S1, '17 S2 ...	'13, '15 S1, '19 ...

Marks in every paper



2010-17	2018-23	2024-26
7.79	8.71	9.00

→ Flat · std. dev. 1.46 · recent questions: 6 MCQ, 10 MSQ, 16 NAT (10 one-mark, 22 two-mark)

1 Belady's anomaly

More frames can mean more faults under FIFO, never under LRU or optimal.

2 Threads vs processes

Threads of one process share globals and the heap, not stacks or registers.

3 Arrival times

In preemptive scheduling, recheck the ready queue at every arrival.

OS · chapters and years asked

Questions per chapter in each of the last 13 papers. A question tagged with two chapters counts in both. The right-hand columns compare with 2010-17.

Chapter	'18	'19	'20	'21 S1	'21 S2	'22	'23	'24 S1	'24 S2	'25 S1	'25 S2	'26 S1	'26 S2	2018-26	2010-17
Process Management I (processes, threads, scheduling)	1	2	2	2	2	1	3	4	3	2	1	2	1	13/13	13/14
Memory Management	1	1	2	1	1	2	2	1	2	2	2	1	2	13/13	12/14
File Systems & Device Management	1	1	1	1		1	1	1	1	2		1	1	11/13	10/14
Process Management II (synchronization)	1	1	1	1	1	1	1	1	1				1	10/13	9/14
Deadlocks	2	1			1	1					1	2		6/13	8/14

Most-asked topics since 2018

TOPIC	QUESTIONS	PAPERS
Process Scheduling	**12**	11/13
Process Synchronization	**10**	10/13
Virtual Memory	**9**	8/13
Memory Management	**9**	8/13
Deadlock Prevention Avoidance Detection	**8**	6/13
Semaphore	**6**	6/13
Page Faults	**5**	5/13
File System	**5**	5/13

STUDY ORDER **Scheduling → Memory management → Synchronization → Deadlock → File systems**

SUBJECT SHEET 3 OF 12

Computer Networks

Tier 1 · Critical | Recent avg 8.17 | Asked in 27/27 papers | Last asked 2026 S2

IN SHORT

- → IP addressing and subnetting (CIDR, masks, host counts) appear in nearly every paper.
- → TCP is the other near-certainty: the handshake, sequence numbers and how the congestion window grows.
- → Data link layer: sliding-window efficiency, CRC, and Ethernet frame-size limits.
- → Routing: distance vector against link state, and the count-to-infinity problem.

RECENT AVG · 2024-26	27-PAPER AVG	LOWEST	HIGHEST
8.17	8.19	6	10
9.6% of 85	2010-2026	'11, '20, '25 S2	'16 S1, '22

Marks in every paper



2010-17	2018-23	2024-26
8.29	8.00	8.17

→ Flat · std. dev. 1.12 · recent questions: 16 MCQ, 8 MSQ, 9 NAT (17 one-mark, 16 two-mark)

1 Usable hosts

Subtract 2 for the network and broadcast addresses unless the question says otherwise.

2 Window sizes

Go-Back-N and Selective Repeat allow different maximum windows for the same sequence bits.

3 Units

Bandwidth is in bits per second and file sizes are usually in bytes. Convert once, at the start.

Networks · chapters and years asked

Questions per chapter in each of the last 13 papers. A question tagged with two chapters counts in both. The right-hand columns compare with 2010-17.

Chapter	'18	'19	'20	'21 S1	'21 S2	'22	'23	'24 S1	'24 S2	'25 S1	'25 S2	'26 S1	'26 S2	2018-26	2010-17
TCP, UDP	3		2	1	1	1	1	2	1	1	2	3	2	12/13	11/14
IP Addresses		2	1	1		2	1	2	2	1	1	1	1	11/13	6/14
Flow Control	2			1	1	2	2	1			1	1	2	9/13	12/14
IP Header	2		1	1				2	2	1	2			7/13	8/14
Application Layer		1	1			1	1	1				1	1	7/13	10/14
Routers & Routing Algorithms					1	2	2	1		1	2			6/13	7/14
Ethernet	1	2			1				2					4/13	5/14
Concept of Layering			1					1		2				3/13	6/14
Error Control				1	1								1	3/13	3/14
Network Security		1												1/13	8/14

Most-asked topics since 2018

TOPIC	QUESTIONS	PAPERS
TCP	15	10/13
Subnetting	10	9/13
Fragmentation	7	7/13
Application Layer Protocols	7	7/13
Sequence Number	6	6/13
Routing	6	5/13
Congestion Control	5	5/13
Network Addressing	5	5/13

STUDY ORDER IP addressing → TCP & UDP → Data link layer → Routing

SUBJECT SHEET 4 OF 12

Computer Organization & Architecture

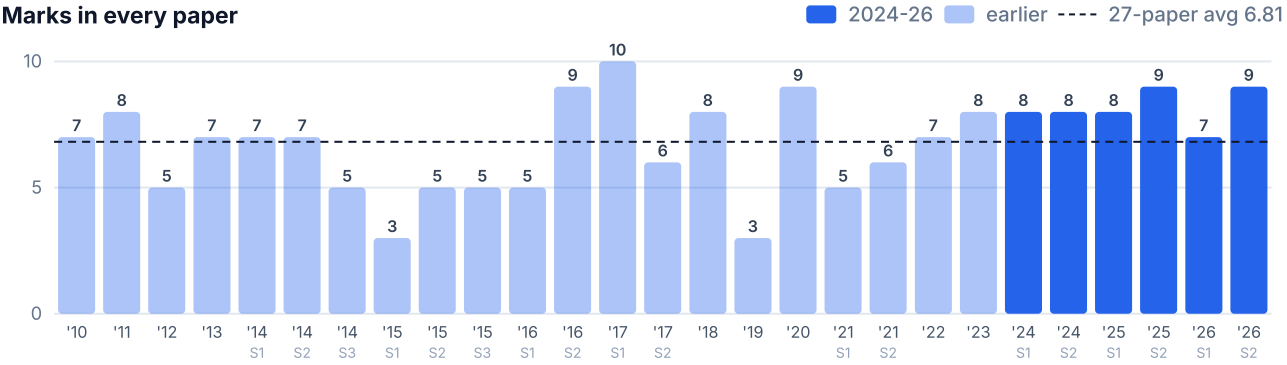
Tier 2 · High | Recent avg 8.17 | Asked in 27/27 papers | Last asked 2026 S2

IN SHORT

- → Cache memory is the single most-asked topic: tag/index/offset bits, mapping, hit ratio and average access time.
- → Pipelining is close behind: speed-up, stalls from hazards and the cycles a branch costs.
- → Memory hierarchy, I/O, DMA and interrupts fill the rest, usually as two-mark numericals.
- → Write each formula with its units before plugging in numbers. Most lost marks are unit slips: ns vs cycles, bits vs bytes.

RECENT AVG · 2024-26	27-PAPER AVG	LOWEST	HIGHEST
8.17	6.81	3	10
9.6% of 85	2010-2026	'15 S1, '19	'17 S1

Marks in every paper



2010-17	2018-23	2024-26
6.36	6.57	8.17

↑ Rising · std. dev. 1.81 · recent questions: 12 MCQ, 6 MSQ, 12 NAT (11 one-mark, 19 two-mark)

1 Hierarchical vs simultaneous access

The access-time formula changes depending on whether levels are checked one after another or in parallel.

2 Word vs byte addressing

This changes the offset bits and every count that follows.

3 Pipeline fill time

n instructions on k stages take k + n - 1 cycles, not n x k.

COA · chapters and years asked

Questions per chapter in each of the last 13 papers. A question tagged with two chapters counts in both. The right-hand columns compare with 2010-17.

Chapter	'18	'19	'20	'21 S1	'21 S2	'22	'23	'24 S1	'24 S2	'25 S1	'25 S2	'26 S1	'26 S2	2018-26	2010-17
Memory Organization	2	2	2	1	2	3	2	2		2	2	2	2	**12/13**	13/14
Pipeline Organization	1		1	1	1	1	1	3	2		1	2	1	**11/13**	12/14
CPU Organization	2		2	1			1		2	2	2	3	1	**9/13**	8/14
Computer Arithmetic			1	1	1	1	1		1		2	1	1	**9/13**	7/14
IO Organization			1		1	1	1	1	1	1			1	**8/13**	3/14
Secondary Memories	1							1	1			1		**4/13**	4/14
Control Unit Design														**0/13**	0/14

Most-asked topics since 2018

TOPIC	QUESTIONS	PAPERS
Cache Memory	**18**	12/13
Pipelining	**12**	11/13
Speedup	**7**	6/13
Instruction Format	**6**	5/13
Interrupts	**5**	5/13
DMA	**5**	5/13
Average Memory Access Time	**4**	4/13
Data Hazards	**4**	4/13

STUDY ORDER **Cache memory → Pipelining → Memory & I/O → CPU organisation**

SUBJECT SHEET 5 OF 12

Engineering Mathematics

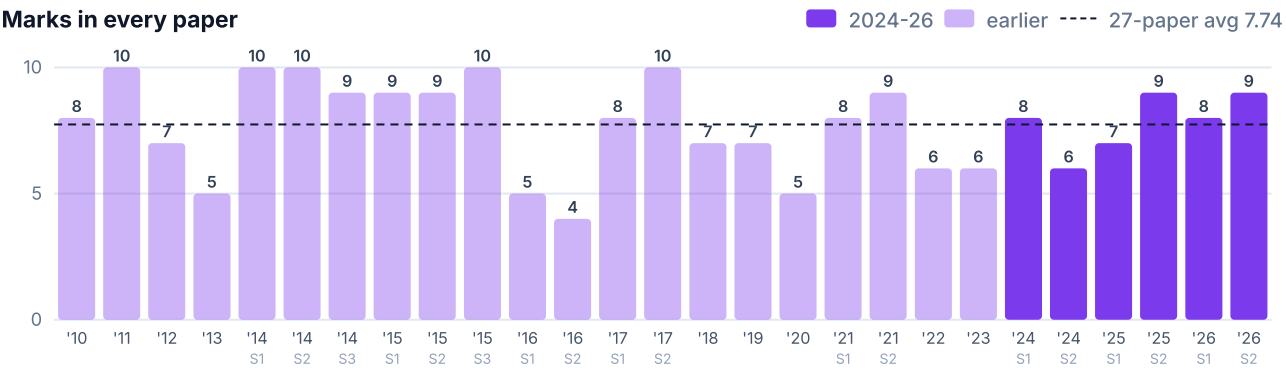
Tier 2 · High | Recent avg 7.83 | Asked in 27/27 papers | Last asked 2026 S2

IN SHORT

- → Probability is the steadiest earner: conditional probability, Bayes, expectation and variance, and the standard distributions.
- → Linear algebra rewards properties over arithmetic: trace = sum of eigenvalues, determinant = product, rank and the number of solutions.
- → Calculus is usually one question: a limit, continuity, a definite integral or a maximum/minimum.
- → Most questions here are numerical answers, so there is no option to check against. Recompute once before moving on.

RECENT AVG · 2024-26	27-PAPER AVG	LOWEST	HIGHEST
7.83	7.74	4	10
9.2% of 85	2010-2026	'16 S2	'11, '14 S1, '14 S2 ...

Marks in every paper



2010-17	2018-23	2024-26
8.14	6.86	7.83

↑ Rising · std. dev. 1.76 · recent questions: 13 MCQ, 8 MSQ, 12 NAT (19 one-mark, 14 two-mark)

1 Claims about matrices

Before accepting a statement about eigenvalues or rank, test it on a 2×2 example.

2 "At least one" probability

Usually quicker as 1 minus the probability of none.

3 Independence vs exclusiveness

Independent events can overlap; mutually exclusive ones cannot both happen.

Engg. Maths · chapters and years asked

Questions per chapter in each of the last 13 papers. A question tagged with two chapters counts in both. The right-hand columns compare with 2010-17.

Chapter	'18	'19	'20	'21 S1	'21 S2	'22	'23	'24 S1	'24 S2	'25 S1	'25 S2	'26 S1	'26 S2	2018-26	2010-17
Probability & Statistics	2	4	1	3	4	1	1	3	2	3	2	2	2	13/13	14/14
Linear Algebra	2	2	1	1	2	4	2	2	2	2	3	2	2	13/13	14/14
Calculus	1	1	1	1	1	1	2	1	1	1	2	2	2	13/13	14/14

Most-asked topics since 2018

TOPIC	QUESTIONS	PAPERS
Probability	**12**	10/13
Eigen Value	**9**	8/13
Determinant	**9**	8/13
Matrix	**9**	4/13
Conditional Probability	**8**	8/13
Definite Integral	**5**	5/13
Expectation	**5**	5/13
System of Equations	**5**	5/13

STUDY ORDER **Probability → Linear algebra → Calculus**

SUBJECT SHEET 6 OF 12

Databases

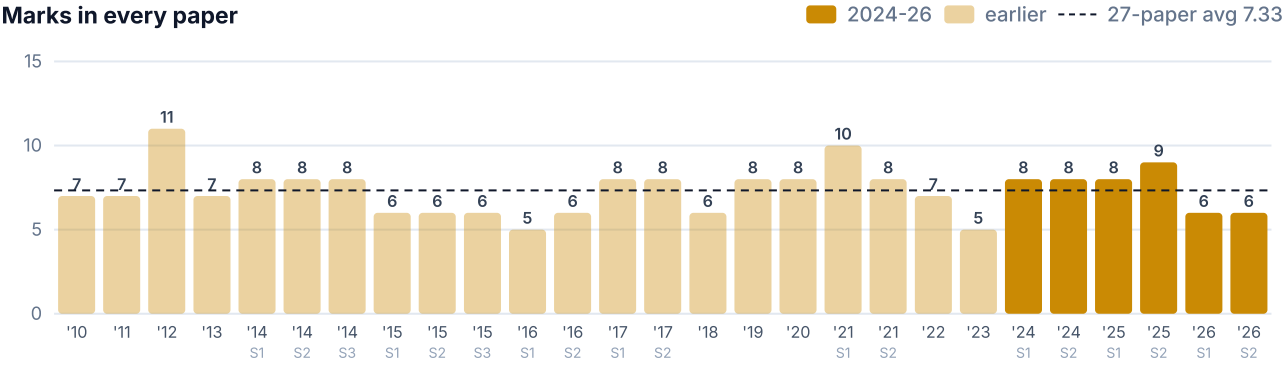
Tier 2 · High | Recent avg 7.50 | Asked in 27/27 papers | Last asked 2026 S2

IN SHORT

- → Functional dependencies and normalization: attribute closures, candidate keys and the highest normal form.
- → SQL and relational algebra: run the query by hand on the small table given.
- → Transactions: conflict serializability with a precedence graph, recoverable schedules and two-phase locking.
- → Indexing: B and B+ tree order, fan-out and the number of levels.

RECENT AVG · 2024-26	27-PAPER AVG	LOWEST	HIGHEST
7.50	7.33	5	11
8.8% of 85	2010-2026	'16 S1, '23	'12

Marks in every paper



2010-17	2018-23	2024-26
7.21	7.43	7.50

→ Flat · std. dev. 1.39 · recent questions: 12 MCQ, 13 MSQ, 5 NAT (15 one-mark, 15 two-mark)

1 NULLs and duplicates

SQL keeps duplicates and treats NULL specially in comparisons and aggregates.

2 Candidate keys

Attributes that never appear on a right-hand side must be in every key. Start from them.

3 View vs conflict serializable

A schedule can be view serializable without being conflict serializable.

DBMS · chapters and years asked

Questions per chapter in each of the last 13 papers. A question tagged with two chapters counts in both. The right-hand columns compare with 2010-17.

Chapter	'18	'19	'20	'21 S1	'21 S2	'22	'23	'24 S1	'24 S2	'25 S1	'25 S2	'26 S1	'26 S2	2018-26	2010-17
Functional Dependencies & Normalization	1	1	1	1	1	2		2	1	1	1	3	1	**12/13**	11/14
Transactions & Concurrency Control		1	1	2	1	1		1	2	1	2		1	**10/13**	11/14
SQL	2	1	1	1	1	1	1	1		1	1			**10/13**	13/14
File Structures & Indexing		1	1	1	1		1	1	1	1	1		1	**10/13**	9/14
Relational Algebra & Relational Calculus	1	1		2		1		1	1	1		1		**8/13**	8/14
ER Diagrams	1		1		1		1	1	1				1	**7/13**	5/14

Most-asked topics since 2018

TOPIC	QUESTIONS	PAPERS
Indexing	**9**	9/13
Functional Dependency	**9**	9/13
SQL	**8**	8/13
Normal Forms	**8**	8/13
Conflict Serializable	**8**	8/13
Relational Algebra	**8**	7/13
Transaction and Concurrency	**8**	7/13
B Tree	**6**	6/13

STUDY ORDER **Normalization → SQL & algebra → Transactions → Indexing**

SUBJECT SHEET 7 OF 12

Algorithms

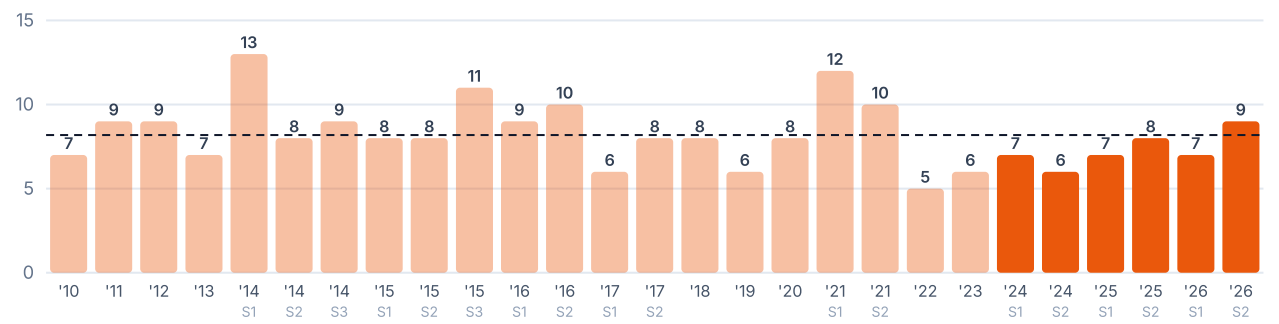
Tier 2 · High | Recent avg 7.33 | Asked in 27/27 papers | Last asked 2026 S2

IN SHORT

- → Asymptotic analysis and recurrences come up every year. Know the Master theorem cold.
- → Graph algorithms: BFS/DFS properties, shortest paths (Dijkstra, Bellman-Ford) and minimum spanning trees (Kruskal, Prim).
- → Sorting and searching: best and worst cases, number of comparisons, stability.
- → Greedy and dynamic programming: write the recurrence, then fill a small table by hand.

RECENT AVG · 2024-26	27-PAPER AVG	LOWEST	HIGHEST
7.33	8.19	5	13
8.6% of 85	2010-2026	'22	'14 S1

Marks in every paper



2010-17	2018-23	2024-26
8.71	7.86	7.33

→ Flat · std. dev. 1.85 · recent questions: 16 MCQ, 5 MSQ, 7 NAT (12 one-mark, 16 two-mark)

1 Negative edges

Dijkstra can fail with negative weights; Bellman-Ford handles them.

2 Unique MSTs

Distinct edge weights guarantee a unique MST. Equal weights may not.

3 Loop counting

Nested loops with changing bounds rarely give n^2. Sum the iterations exactly.

Algorithms · chapters and years asked

Questions per chapter in each of the last 13 papers. A question tagged with two chapters counts in both. The right-hand columns compare with 2010-17.

Chapter	'18	'19	'20	'21 S1	'21 S2	'22	'23	'24 S1	'24 S2	'25 S1	'25 S2	'26 S1	'26 S2	2018-26	2010-17
Algorithm Analysis & Asymptotic Notations	1	1	6	4	1	3	3	3	1	1		1	3	**12/13**	14/14
Greedy Method	2	1	4	2	3	1			2	2	1	2	2	**11/13**	10/14
Miscellaneous Topics	1		1	3	1		2	2	1	3	3	3	2	**11/13**	13/14
Dynamic Programming	2	1		1	1				1			1	1	**7/13**	7/14
Divide & Conquer		1			1						1		1	**4/13**	9/14
Optimal Binary Search Trees														**0/13**	0/14

Most-asked topics since 2018

TOPIC	QUESTIONS	PAPERS
Minimum Spanning Tree	**14**	11/13
Time Complexity	**13**	8/13
Recurrence Relation	**9**	9/13
Shortest Path	**8**	7/13
Graph Search	**8**	6/13
Depth First Search	**7**	6/13
Dynamic Programming	**7**	6/13
Asymptotic Notations	**6**	5/13

STUDY ORDER **Complexity & recurrences → Graph algorithms → Sorting → Greedy & DP**

SUBJECT SHEET 8 OF 12

Digital Logic

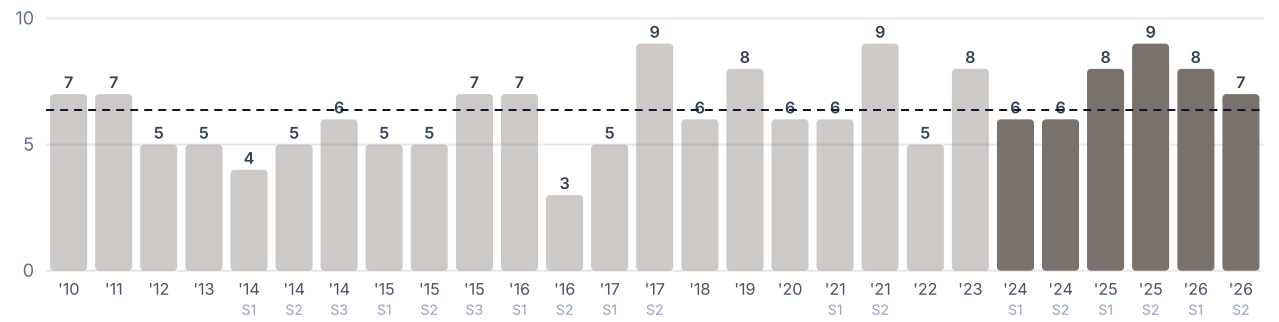
Tier 3 · Steady | Recent avg 7.33 | Asked in 27/27 papers | Last asked 2026 S2

IN SHORT

- → Boolean algebra and K-maps: minimal sum of products, prime and essential prime implicants.
- → Number systems: two's complement ranges and overflow, base conversion and IEEE 754 floating point.
- → Combinational circuits: multiplexers implementing functions, adders and decoders.
- → Sequential circuits: flip-flop behaviour, counters and state tables.

RECENT AVG · 2024-26	27-PAPER AVG	LOWEST	HIGHEST
7.33	6.37	3	9
8.6% of 85	2010-2026	'16 S2	'17 S2, '21 S2, '25 S2

Marks in every paper



2010-17	2018-23	2024-26
5.71	6.86	7.33

→ Flat · std. dev. 1.54 · recent questions: 7 MCQ, 15 MSQ, 7 NAT (14 one-mark, 15 two-mark)

1 Overflow

Adding two numbers of the same sign can overflow; adding opposite signs never does.

2 IEEE 754 bias

Single precision uses bias 127. Denormals have no implicit leading 1.

3 Don't-cares

Use them to grow groups, but never form a group of don't-cares alone.

Digital Logic · chapters and years asked

Questions per chapter in each of the last 13 papers. A question tagged with two chapters counts in both. The right-hand columns compare with 2010-17.

Chapter	'18	'19	'20	'21 S1	'21 S2	'22	'23	'24 S1	'24 S2	'25 S1	'25 S2	'26 S1	'26 S2	2018-26	2010-17
Number Systems	1	2	1	2	3	2	2	1	2	1	2	2	2	**13/13**	10/14
Boolean Algebra	1	3	1	1				1	2	1	2	1	1	**10/13**	14/14
Combinational Circuits		2	3		1	1	3	1	1		1		1	**9/13**	11/14
K-Maps	1	1			1			1		1	1	2	1	**8/13**	8/14
Sequential Circuits	1			1	1		2			2	1	1		**7/13**	9/14

Most-asked topics since 2018

TOPIC	QUESTIONS	PAPERS
Boolean Algebra	**10**	9/13
IEEE Representation	**9**	9/13
Floating Point Representation	**8**	8/13
K Map	**8**	7/13
Number Representation	**8**	7/13
Multiplexer	**7**	6/13
Sequential Circuit	**6**	5/13
Flip Flop	**4**	4/13

STUDY ORDER **Boolean algebra & K-maps → Number systems → Combinational → Sequential**

SUBJECT SHEET 9 OF 12

Theory of Computation

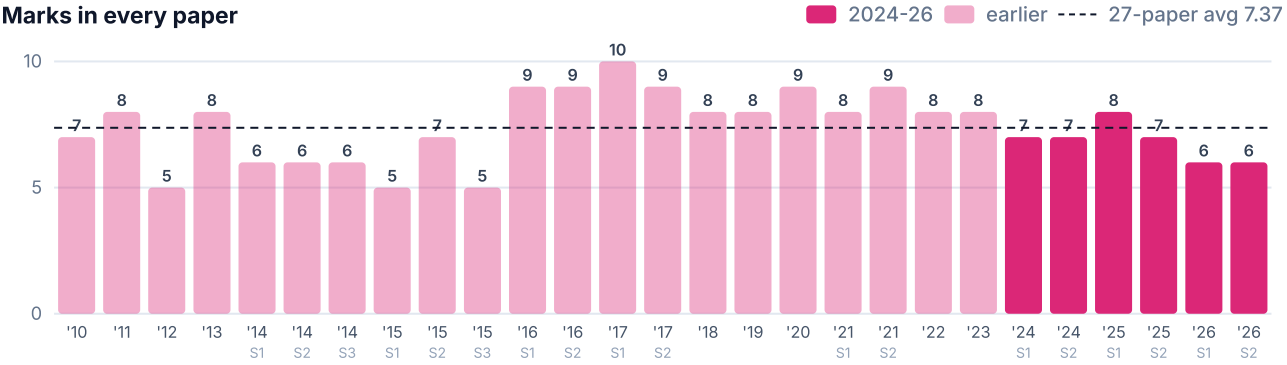
Tier 3 · Steady | Recent avg 6.83 | Asked in 27/27 papers | Last asked 2026 S2

IN SHORT

- → Regular languages dominate: DFA construction and minimisation, regular expressions and closure properties.
- → Classifying a language as regular, context-free or neither is a recurring question. Ask what has to be counted.
- → Context-free languages: grammars, PDAs and which operations keep a language context-free.
- → Decidability: know the standard decidable and undecidable problems, and Rice's theorem.

RECENT AVG · 2024-26	27-PAPER AVG	LOWEST	HIGHEST
6.83	7.37	5	10
8.0% of 85	2010-2026	'12, '15 S1, '15 S3	'17 S1

Marks in every paper



2010-17	2018-23	2024-26
7.14	8.29	6.83

↓ Falling · std. dev. 1.36 · recent questions: 8 MCQ, 14 MSQ, 4 NAT (11 one-mark, 15 two-mark)

1 Closure under intersection

CFLs are not closed under intersection, but CFL intersected with a regular language is a CFL.

2 Finite languages

Every finite language is regular, however strange it looks.

3 Minimal DFA states

Count the dead state if the question wants a complete DFA.

ToC · chapters and years asked

Questions per chapter in each of the last 13 papers. A question tagged with two chapters counts in both. The right-hand columns compare with 2010-17.

Chapter	'18	'19	'20	'21 S1	'21 S2	'22	'23	'24 S1	'24 S2	'25 S1	'25 S2	'26 S1	'26 S2	2018-26	2010-17
RL, FA, RE & RG	3	3	3	1	4	1	3	3	3	4	2	2	1	13/13	14/14
CFL & PDA	2	2	3	2	3	3	3	2	1	2	3	3	2	13/13	13/14
CSL, RS, RES, LBA & TM	2	2	2	1	2	4				2	1		2	9/13	13/14
Undecidability	1		1	2	1	1					1		1	7/13	9/14

Most-asked topics since 2018

TOPIC	QUESTIONS	PAPERS
Regular Language	12	11/13
Closure Property	11	10/13
Finite Automata	11	9/13
Identify Class Language	10	8/13
Context Free Language	9	8/13
Regular Expression	9	7/13
Minimal State Automata	8	8/13
Decidability	8	7/13

STUDY ORDER **Regular languages → Context-free languages → Turing machines & decidability**

SUBJECT SHEET 10 OF 12

Compiler Design

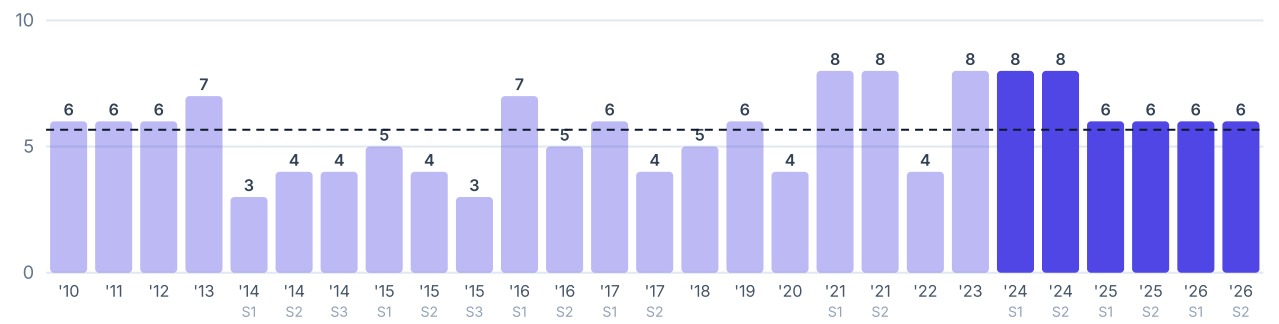
Tier 3 · Steady | Recent avg 6.67 | Asked in 27/27 papers | Last asked 2026 S2

IN SHORT

- → Parsing is asked every year: FIRST and FOLLOW sets, LL(1) tables, and LR(0), SLR, LALR and CLR conflicts.
- → Syntax-directed translation: evaluate attributes on a parse tree; know S- vs L-attributed.
- → Code optimisation: basic blocks, DAGs, live variables and the minimum number of registers.
- → Lexical analysis: count tokens and know which errors each phase catches.

RECENT AVG · 2024-26	27-PAPER AVG	LOWEST	HIGHEST
6.67	5.67	3	8
7.8% of 85	2010-2026	'14 S1, '15 S3	'21 S1, '21 S2, '23 ...

Marks in every paper



2010-17	2018-23	2024-26
5.00	6.14	6.67

→ Flat · std. dev. 1.54 · recent questions: 15 MCQ, 7 MSQ, 4 NAT (12 one-mark, 14 two-mark)

1 Parser power

LR(0) is weaker than SLR, then LALR, then CLR. Every LL(1) grammar is also LR(1).

2 Epsilon productions

They pull FOLLOW sets into FIRST calculations and change LL(1) tables.

3 Error phases

An unterminated string is a lexical error; a type mismatch is semantic.

Compiler · chapters and years asked

Questions per chapter in each of the last 13 papers. A question tagged with two chapters counts in both. The right-hand columns compare with 2010-17.

Chapter	'18	'19	'20	'21 S1	'21 S2	'22	'23	'24 S1	'24 S2	'25 S1	'25 S2	'26 S1	'26 S2	2018-26	2010-17
Parsing Techniques	1	3	2	2	1	2		3	2	1	2	2	2	**12/13**	12/14
Syntax Directed Translation		1	1	2	1	1	1	3	2	1	2	1		**11/13**	6/14
Lexical Analysis	2		1	1	1	1	2		1	1		1	1	**10/13**	7/14
Code Optimization & Runtime Environment	2		1	2	2		2	1		2		2	3	**9/13**	13/14

Most-asked topics since 2018

TOPIC	QUESTIONS	PAPERS
Parsing	**10**	8/13
Syntax Directed Translation	**9**	9/13
LR Parser	**9**	7/13
Compilation Phases	**6**	6/13
First and Follow	**5**	5/13
Intermediate Code	**5**	5/13
Lexical Analysis	**4**	4/13
Runtime Environment	**4**	4/13

STUDY ORDER **Parsing → Syntax-directed translation → Optimisation → Lexical analysis**

SUBJECT SHEET 11 OF 12

Discrete Mathematics

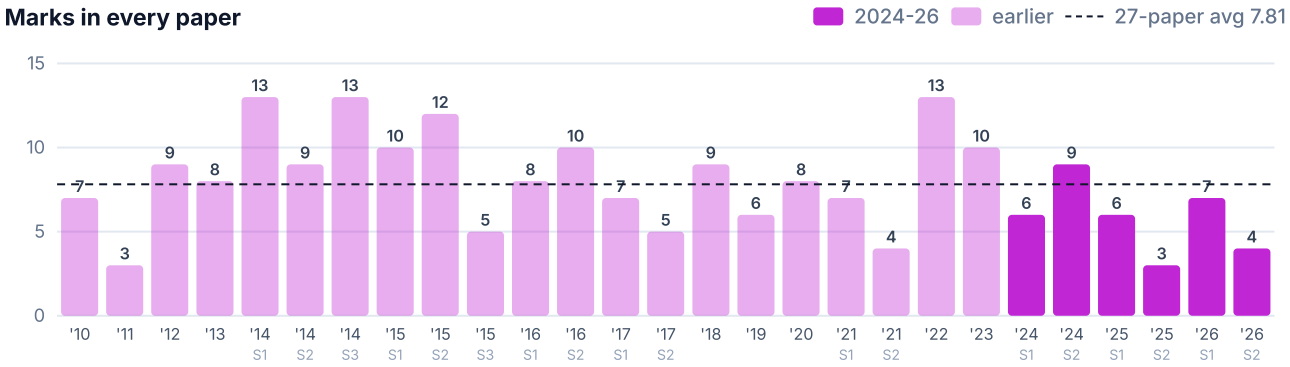
Tier 3 · Steady | Recent avg 5.83 | Asked in 27/27 papers | Last asked 2026 S2

IN SHORT

- → Counting is the core: permutations, combinations, pigeonhole, inclusion-exclusion and recurrences.
- → Graph theory appears almost every year: degree sums, connectivity, colouring, matchings and Euler's formula for planar graphs.
- → Logic questions test translation: turn English into quantifiers carefully, then check validity.
- → Relations and groups: equivalence relations, partial orders and lattices, and Lagrange's theorem for subgroup and element orders.

RECENT AVG · 2024-26	27-PAPER AVG	LOWEST	HIGHEST
5.83	7.81	3	13
6.9% of 85	2010-2026	'11, '25 S2	'14 S1, '14 S3, '22

Marks in every paper



2010-17	2018-23	2024-26
8.50	8.14	5.83

↓ Falling · std. dev. 2.87 · recent questions: 7 MCQ, 9 MSQ, 7 NAT (11 one-mark, 12 two-mark)

1 Order of quantifiers

"For all x there is a y" is not the same as "there is a y for all x".

2 Labelled vs unlabelled

Check whether the question counts distinct objects or distinct shapes.

3 Implications

An option may only hold one way. Check both directions before calling it "if and only if".

Discrete Maths · chapters and years asked

Questions per chapter in each of the last 13 papers. A question tagged with two chapters counts in both. The right-hand columns compare with 2010-17.

Chapter	'18	'19	'20	'21 S1	'21 S2	'22	'23	'24 S1	'24 S2	'25 S1	'25 S2	'26 S1	'26 S2	2018-26	2010-17
Set Theory & Algebra	2	5	2	2	2	1	2	2	2	2	1		1	**12/13**	12/14
Combinatorics	2	2	2	1	1	4	3	1	1	1		2		**11/13**	12/14
Graph Theory	2	1	1	3		5	2	3	4			3	1	**10/13**	12/14
Mathematical Logic	1	1	1	1	1		1		1	1	1		1	**10/13**	14/14

Most-asked topics since 2018

TOPIC	QUESTIONS	PAPERS
Counting	**13**	9/13
Graph Theory	**13**	8/13
Group Theory	**9**	9/13
Graph Coloring	**7**	7/13
First Order Logic	**7**	7/13
Combinatory	**7**	7/13
Relations	**6**	6/13
Set Theory	**5**	4/13

STUDY ORDER **Combinatorics → Graph theory → Logic → Set theory & algebra**

SUBJECT SHEET 12 OF 12

General Aptitude

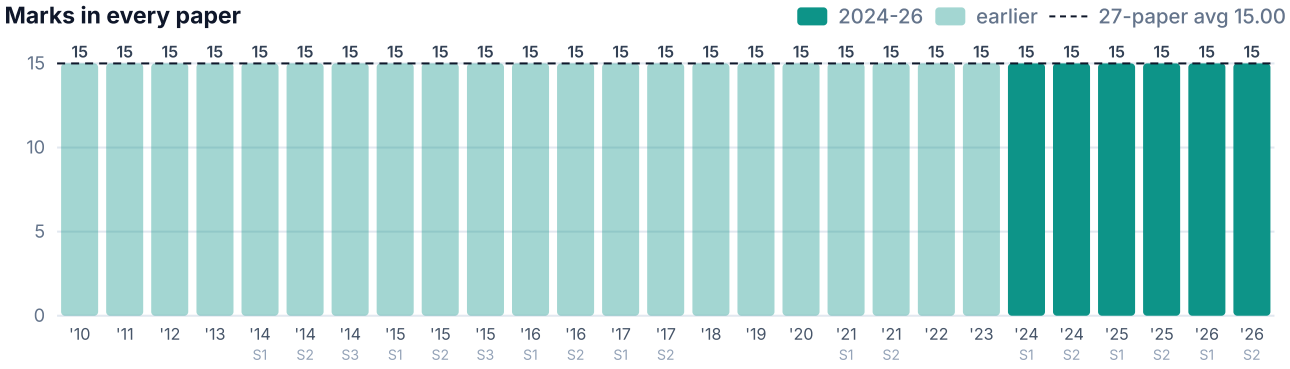
Fixed 15 marks | Recent avg 15.00 | Asked in 27/27 papers | Last asked 2026 S2

IN SHORT

- → 15 marks in every paper since 2010: 5 one-mark and 5 two-mark questions.
- → Verbal ability: grammar, word meaning and short passages.
- → Quantitative aptitude: percentages, ratios, averages, counting and probability.
- → Reasoning, data interpretation and spatial puzzles (paper folding, rotation, patterns), which have been in every paper since 2021.

RECENT AVG · 2024-26	27-PAPER AVG	LOWEST	HIGHEST
15.00	15.00	15	15
out of 15	2010-2026	'10, '11, '12 ...	'10, '11, '12 ...

Marks in every paper



2010-17	2018-23	2024-26
15.00	15.00	15.00

→ Fixed · std. dev. 0.00 · recent questions: 60 MCQ, 0 MSQ, 0 NAT (30 one-mark, 30 two-mark)

1 Time

Aim for under 25 minutes on this section. It is the easiest 15 marks on the paper.

2 Passage questions

Choose what the passage supports, not what is true in general.

3 Spatial questions

Sketch the figure. Mental rotation is where most errors happen.

Aptitude · chapters and years asked

Questions per chapter in each of the last 13 papers. A question tagged with two chapters counts in both. The right-hand columns compare with 2010-17.

Chapter	'18	'19	'20	'21 S1	'21 S2	'22	'23	'24 S1	'24 S2	'25 S1	'25 S2	'26 S1	'26 S2	2018-26	2010-17
Quantitative Aptitude	7	4	4	5	4	6	4	6	5	4	3	4	4	13/13	14/14
Verbal Ability	2	3	3	2	2	1	3	2	2	2	2	1	2	13/13	14/14
Reading & Logical Reasoning	1	3	4	2	2	4	4		2	3	4	4	3	12/13	14/14
Data Interpretation & Spatial Reasoning		1	1	1	4	2	2	3	2	2	3	2	4	12/13	10/14

Most-asked topics since 2018

TOPIC	QUESTIONS	PAPERS
Logical Reasoning	**11**	8/13
Logical Inference	**10**	7/13
Verbal Analogy	**9**	9/13
Spatial Ability	**9**	8/13
Passage Reading	**8**	7/13
Probability	**7**	6/13
Most Appropriate Word	**6**	6/13
Percentage	**6**	6/13

STUDY ORDER **Quantitative → Verbal → Reasoning & data interpretation → Spatial**

06 How reliable is each chapter?

Every chapter in the playbook, grouped by how many of the last 13 papers (2018-2026) asked at least one question from it. The number on each chip is that count. **12 chapters** appeared in every one of them.

13/13
Every paper
12 chapters

Never skipped since 2018. As close to guaranteed as GATE gets.

- Quantitative Aptitude 13
- Programming Languages 13
- RL, FA, RE & RG 13
- CFL & PDA 13
- Probability & Statistics 13
- Trees 13
- Linear Algebra 13
- Verbal Ability 13
- Process Management I (processes, threads, scheduling) 13
- Number Systems 13
- Memory Management 13
- Calculus 13

11-12/13
Almost always
15 chapters

Missing from at most two papers.

- Reading & Logical Reasoning 12
- Algorithm Analysis & Asymptotic Notations 12
- Data Interpretation & Spatial Reasoning 12
- Set Theory & Algebra 12
- Memory Organization 12
- Parsing Techniques 12
- TCP, UDP 12
- Functional Dependencies & Normalization 12
- Greedy Method 11
- Miscellaneous Topics 11
- Combinatorics 11
- Syntax Directed Translation 11
- Pipeline Organization 11
- IP Addresses 11
- File Systems & Device Management 11

7-10/13
Frequent
27 chapters

In most papers, but skipped now and then.

- Graph Theory 10
- Boolean Algebra 10
- Transactions & Concurrency Control 10
- Lexical Analysis 10
- SQL 10
- Mathematical Logic 10
- File Structures & Indexing 10
- Process Management II (synchronization) 10
- CSL, RS, RES, LBA & TM 9
- Code Optimization & Runtime Environment 9
- CPU Organization 9
- Combinational Circuits 9
- Flow Control 9
- Computer Arithmetic 9
- Arrays 8
- Graphs 8
- Relational Algebra & Relational Calculus 8
- K-Maps 8
- IO Organization 8
- Linked Lists 8
- IP Header 7
- Sequential Circuits 7
- Stacks & Queues 7
- Dynamic Programming 7
- Undecidability 7
- Application Layer 7
- ER Diagrams 7

4-6/13
Occasional
6 chapters

Roughly one paper in three.

- Routers & Routing Algorithms 6
- Deadlocks 6
- Hashing 6
- Ethernet 4
- Secondary Memories 4
- Divide & Conquer 4

0-3/13
Rare
5 chapters

Low yield. Learn them after everything else.

- Concept of Layering 3
- Error Control 3
- Network Security 1
- Control Unit Design 0
- Optimal Binary Search Trees 0

How to read this. Chapters follow the ExamPrep book-chapter map used across the site. A chapter can be "every paper" with only one question each time, so pair this page with the chapter grids in Section 05 to see how many questions it brings.

07 Exam-day checklist

Habits that save marks. Read them the night before, then run them on autopilot during the paper.

- [ ] Read every option of a "which is NOT / FALSE" question before marking one.
- [ ] MSQ and NAT have no negative marking: never leave them blank.
- [ ] MCQs cost a third of their marks when wrong. Guess only after removing at least one option.
- [ ] For any C code, write a table of variable values and trace line by line.
- [ ] For a claim about matrices, graphs or sets, try it on the smallest example you can build.
- [ ] Draw the Gantt chart, the pipeline diagram or the timeline. Don't compute those in your head.
- [ ] Write units next to every number in a numerical question.
- [ ] An "if and only if" option must hold in both directions. Check the reverse.
- [ ] A question that mixes two subjects (say, networks and OS) needs extra time. Budget for it.
- [ ] Finish General Aptitude in about 25 minutes, then do the one-mark technical block.
- [ ] Mark long two-mark questions and come back after the quick ones.
- [ ] Leave 10 minutes at the end to recheck NAT answers and rounding.

Before you start

65 questions · 100 marks · 3 hours. 15 aptitude + 85 technical. Answer every MSQ and NAT. Guess on MCQs only after removing an option.

08 About the data

Dataset

27 official GATE CS papers: 2010, 2011, 2012, 2013, 2014 S1, 2014 S2, 2014 S3, 2015 S1, 2015 S2, 2015 S3, 2016 S1, 2016 S2, 2017 S1, 2017 S2, 2018, 2019, 2020, 2021 S1, 2021 S2, 2022, 2023, 2024 S1, 2024 S2, 2025 S1, 2025 S2, 2026 S1, 2026 S2. 1,755 questions, each tagged with its subject, its book chapter(s) and finer GATE Overflow topics in the ExamPrep question bank. Every paper totals 100 marks.

How the numbers are computed

- **Marks** for a subject are the marks of the questions whose main subject it is.
- **Recent average** is the mean over the 6 papers of 2024-2026 (2024 S1, 2024 S2, 2025 S1, 2025 S2, 2026 S1, 2026 S2).
- **Eras:** 2010-17 (14 papers), 2018-23 (7), 2024-26 (6).
- **Standard deviation** is taken over all 27 papers.
- **Tiers:** the top 3 subjects by recent average are Tier 1, the next 4 Tier 2, the rest Tier 3.
- **Chapter counts** use the book-chapter tags. A question tagged with two chapters counts in both.

What is excluded

Papers from 2007-2009 (a different format: no aptitude section and 150 marks in 2007-08). Software engineering and web technology questions, last asked in 2016, appear in Section 02 but are left out of every average.

Keep it current

The playbook is generated from the question bank. When the 2027 papers are added, rebuild it and every number, chart and ranking updates.

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